Operating instructions

V 1.0 English

QiTech Extruder

QiTech Industries GmbH
Landwehrstraße 55
64293 Darmstadt

EC Declaration of Conformity

According to 2006/42/EC, Annex II, No. 1 A

The content of the EC Declaration of Conformity corresponds to Annex II, No. 1 A of Directive 2006/42/EC.

Manufacturer of the machine:

QiTech Industries GmbH
Landwehrstraße 55
64293 Darmstadt

Representative of the manufacturer:

Milan von dem Bussche
Landwehrstraße 55
64293 Darmstadt

Person authorised to compile the technical file:

Milan von dem Bussche
Landwehrstraße 55
64293 Darmstadt

General designation	Filament extrusion system
Type	Filament extrusion system
Function	Extrusion of thermoplastic filament from granulate
Model	QiTech Extruder
Trade name	QiTech Extruder
Serial number	

We hereby declare that the above-mentioned machine is in conformity with all relevant provisions of the EC Machinery Directive 2006/42/EC. The machine is also in compliance with all relevant provisions of the following EC Directives:

- 2014/30/EU Electromagnetic Compatibility / European EMC Directive
- 2014/35/EU Low Voltage Directive (protection objectives only)
- 2011/65/EU ROHS Directives (ROHS 2)

The following harmonized standards (or parts of these standards) have been applied:

- DIN EN ISO 12100
- DIN EN ISO 20607

Darmstadt, 08.09.2026



Milan von dem Bussche

Read before commissioning!

These operating instructions are intended to enable all people working on the machine to carry out their work safely and correctly. Every person who operates, maintains or repairs the machine must have read the operating instructions, particularly the safety regulations, before commissioning. These operating instructions must always be available on the machine. The operator must ensure that all information necessary for the safe operation of the machine is made available to the operating personnel. If the operating instructions are available in electronic form, the operator must ensure that the operating personnel are able to view the operating instructions. The original version of these operating instructions is written in English. We ask you to read through the relevant instructions carefully and to observe them thoroughly. After unpacking the machine, check for any damage that may have occurred during transport.

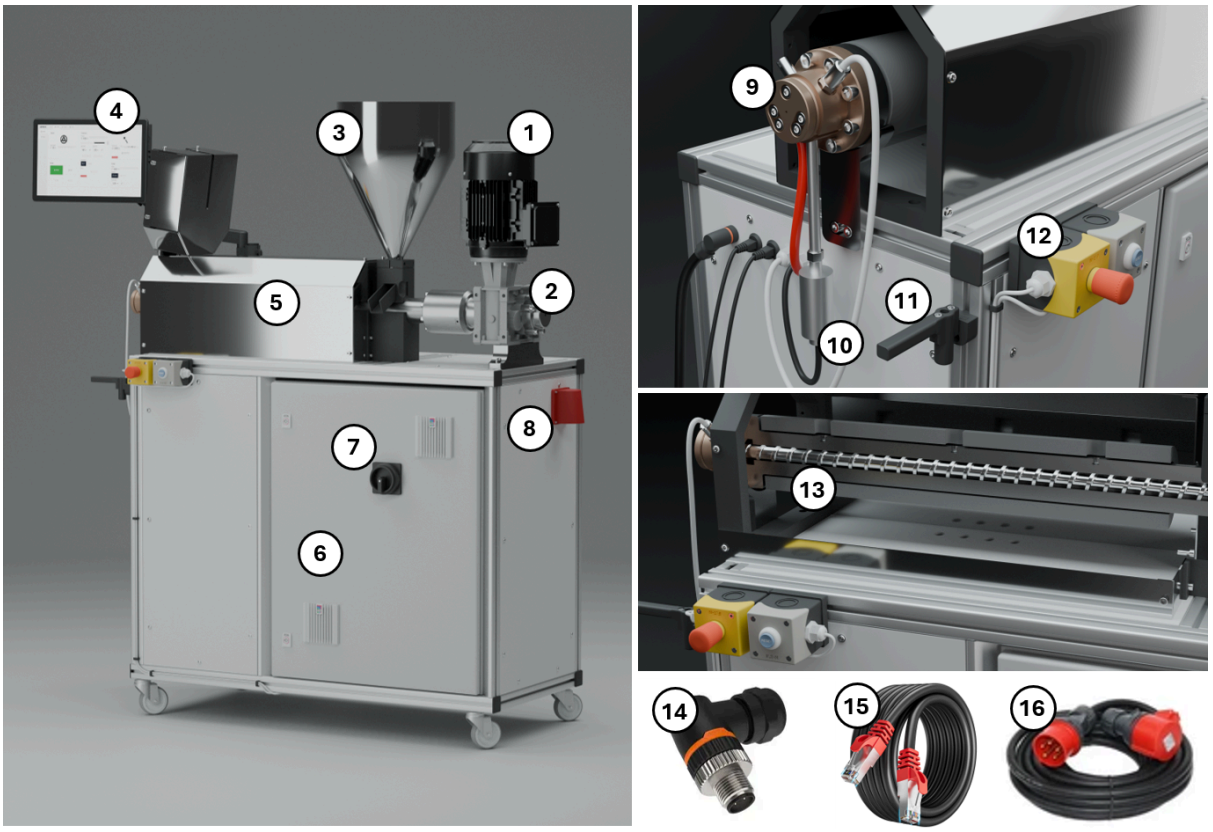
NOTICE: Before using the machine for the first time, please inform yourself about the national regulations (e.g. occupational safety, environmental restrictions). Reproduction and reprinting in any form, even in extracts, requires the written authorization of QiTech Industries GmbH.

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1. Machine description

1.1 Component overview



Item number	Description
1	Drive motor
2	Gearbox and coupling housing
3	Feed hopper
4	Control display
5	Heat barrel
6	Control cabinet
7	Main switch

8	Three-phase power connection (CEE, 400 V)
9	Nozzle plate and nozzle head
10	Pressure sensor
11	Die head holder for screw changeover
12	Emergency stop button and reset button
13	Extruder screw
14	Power-supply-connector (5 pin M12)
15	LAN-cable
16	CEE power cable

The design of the main control elements depends on how the QiTech Extruder is operated.

Because extrusion involves an increased hazard from hot surfaces, pressurised melt and the rotating screw, the QiTech Extruder is always equipped with two separate control elements. A grey rotary main switch energises and de-energises the machine, and a separate red emergency stop button in a yellow housing serves as the emergency switching-off device.

In standalone operation, both elements are located on the QiTech Extruder itself. The emergency stop button switches off the machine immediately and needs to be released manually before the extruder can be restarted.

When the QiTech Extruder is operated as part of a process line together with the QiTech Winder and the QiTech Aquapath or QiTech Airpath, a central emergency stop is provided at the cooling unit for easy access from the front of the process line. It switches off the entire process line and substitutes the emergency stop on the QiTech Extruder. The main switch on the QiTech Extruder retains the same function, due to the power connection it switches off the whole process line.

1.2 Technical specifications

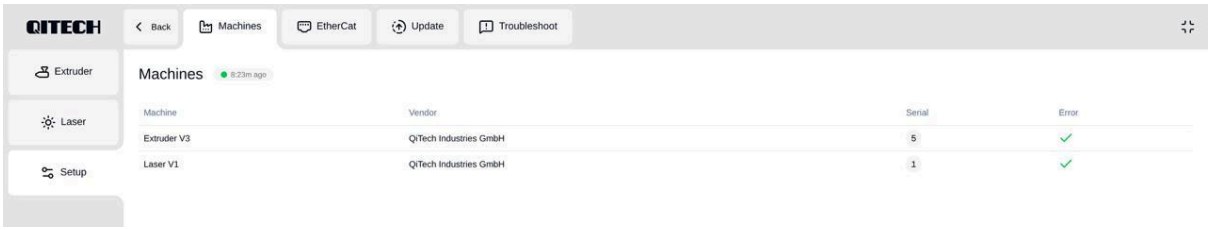
Dimensions and weight	
Length x Width x Height	2870mm x 600mm x 1120mm
Weight	146 kg
Screw dimensions (Lenght x Diameter)	897 mm x 25 mm
Electrical specifications	
Rated voltage	400 V AC, 3~
Rated frequency	50 Hz
Rated apparent power	4,0 kVA
Rated current	5,77 A
Process parameters	
Number of heating zones	3 barrel zones + 1 nozzle zone
Heating power per zone	900 W barrel zones, 150 W nozzle zone
Maximum temperature per zone	350 °C
Maximum nozzle pressure	300 bar
Maximum pressure sensor measuring range	350 bar
Maximum screw speed	50 rpm
Barrel volume	15 l
Noise emissions	
Workplace-related emission value (L_pA)	< 50 dB

1.3 Controls and interface

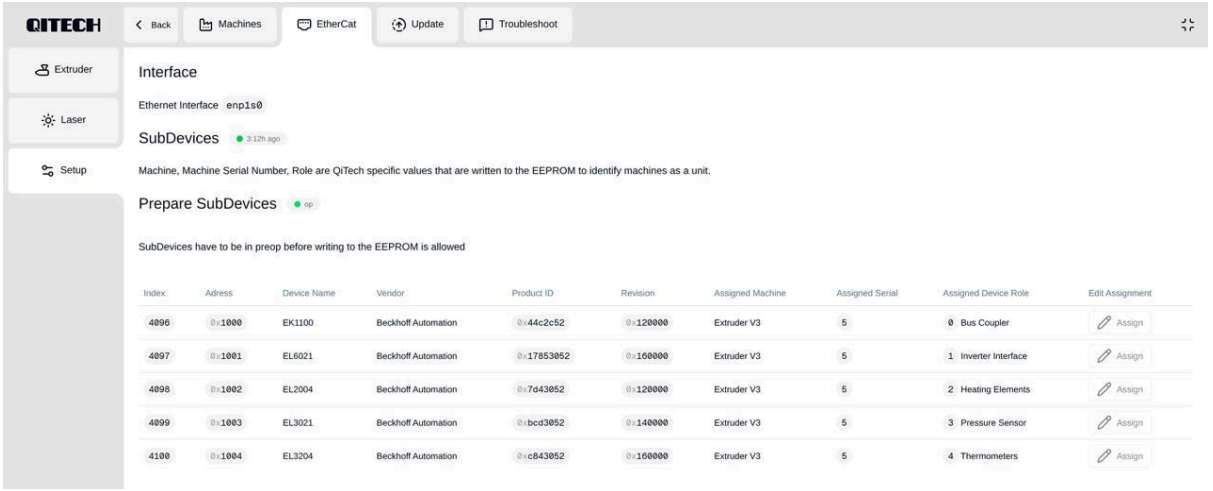
1.3.1 General controls and interface

The QiTech Extruder is operated entirely via the QiTech Control software. The machine is accessible on the left-hand sidebar, which makes it possible to control multiple machines simultaneously.

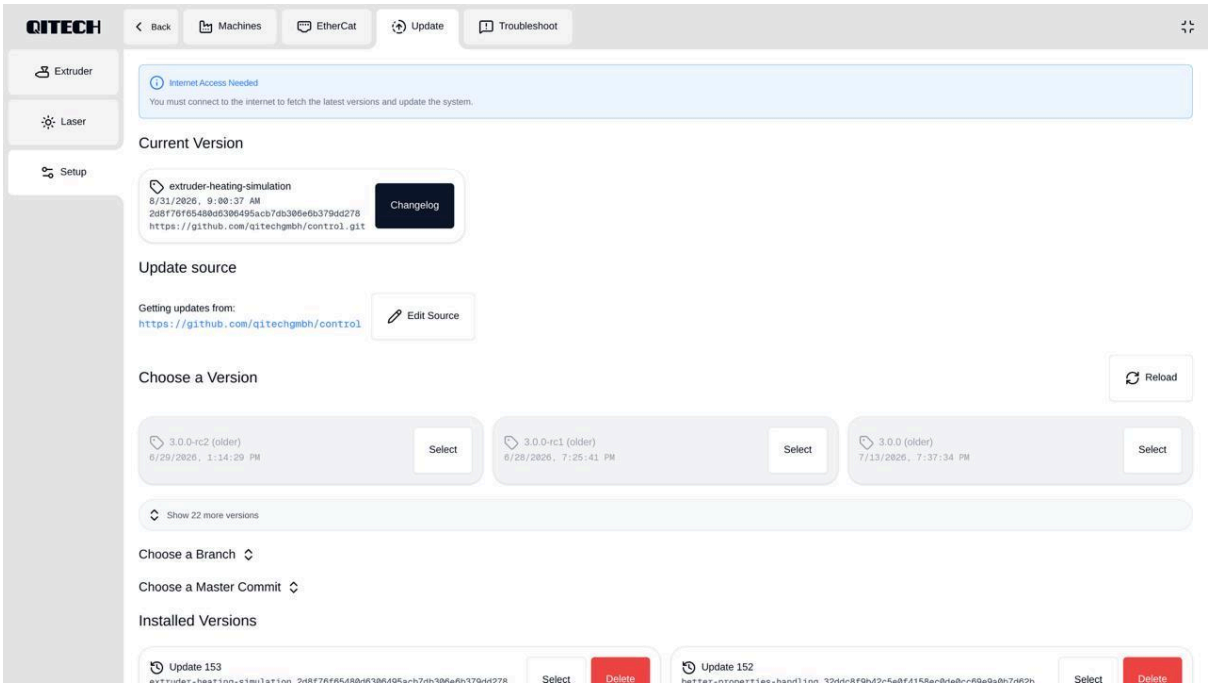
The sidebar on the left switches between the views of the connected machine. *Set Up* provides access to machine identification, EtherCAT device configuration, software updates and system troubleshooting functions.



Machines	
Machine	Shows the name and version of the configured machine
Vendor	Shows the manufacturer of the respective machine
Serial	Displays the assigned machine serial number
Error Status	Indicates whether an error is currently detected for the machine

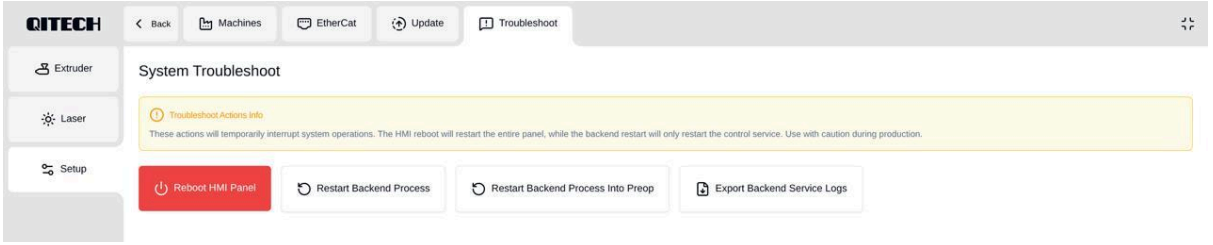


Ethercat	
Ethernet Interface	Shows the Ethernet interface currently used for EtherCAT communication
SubDevices	Displays all detected EtherCAT subdevices connected to the system
Index	Shows the internal index of the respective EtherCAT subdevice
Address	Shows the EtherCAT address assigned to the subdevice
Device Name	Displays the detected EtherCAT device type
Vendor	Shows the manufacturer of the EtherCAT device
Product ID	Displays the EtherCAT product identification number
Revision	Displays the hardware/device revision
Assigned Machine	Shows the machine to which the subdevice is assigned
Assigned Serial	Shows the serial number of the assigned machine
Assigned Device Role	Shows the function of the subdevice within the machine
Edit Assignment	Opens the assignment settings for the selected subdevice



Update	
Current Version	Displays the software version currently installed and running
Changelog	Opens the changelog of the current software version
Update Source	Shows the repository or source from which software updates are retrieved; Edit source if software update source must be changed
Choose a Version	Displays available software versions that can be installed; Select the corresponding software version for installation
Reload	Refreshes the list of available software versions
Choose a Branch	Allows a software branch to be selected manually
Choose a Master Commit	Allows a specific master commit to be selected manually

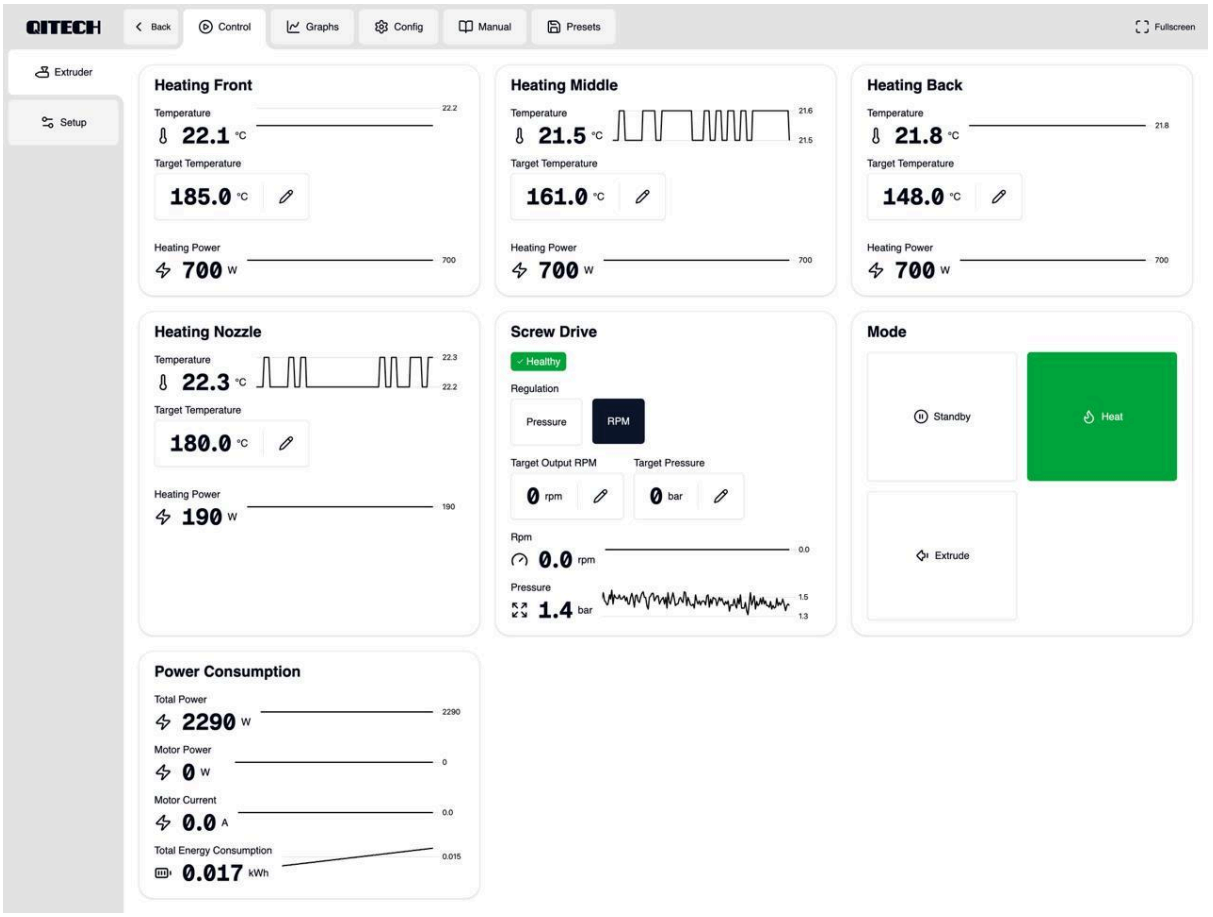
Select Installed Version	Displays software versions already stored on the system; Select or delete a previously installed software version



Troubleshoot	
Reboot HMI Panel	Reboots the complete HMI panel and restarts the user interface
Restart Backend Process	Restarts the backend control process without rebooting the complete HMI panel
Restart Backend Process Into PREOP	Restarts the backend process and places the EtherCAT system into PREOP state
Export Backend Service Logs	Exports backend service log files for troubleshooting and diagnostic purposes

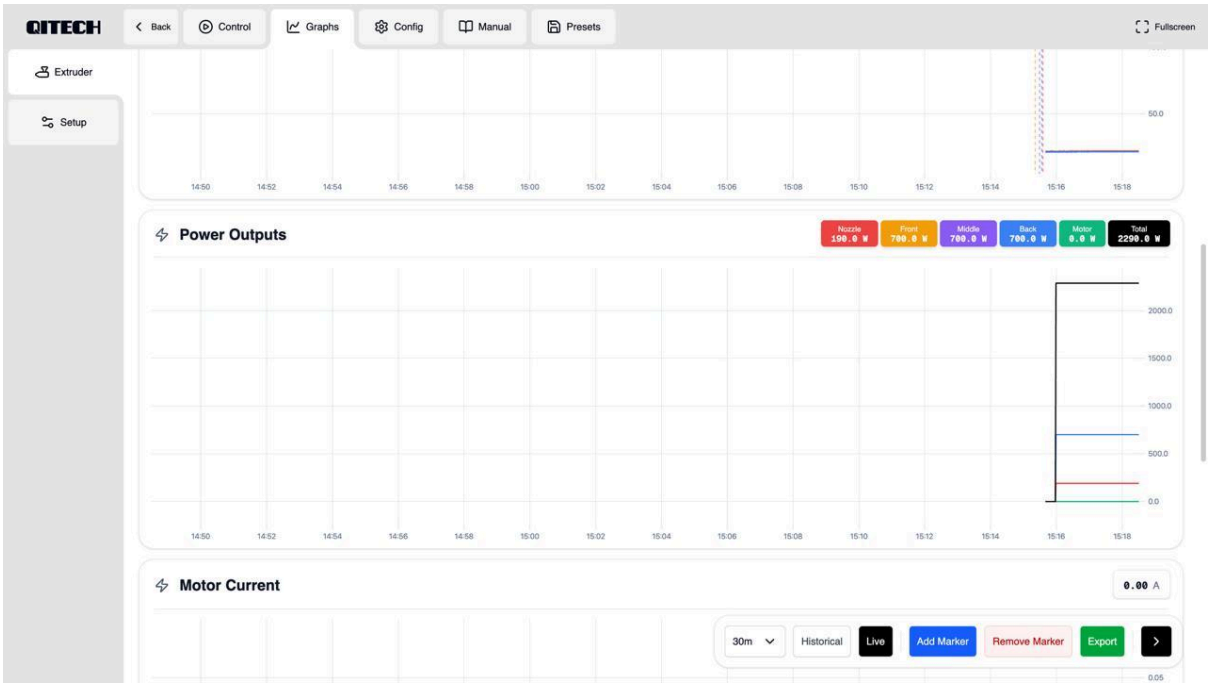
1.3.2 Machine-specific controls and interface

The navigation bar at the top of the screen provides access to the main areas of the QiTech Control software. The *Control* view is the primary operating view, displaying all real-time machine parameters. The *Graphs* view shows process data graphically over time and allows data to be exported in Excel format. The *Config* view provides access to the machine configuration settings. The *Manual* view contains this operating manual, so that all instructions remain available directly at the machine. The *Presets* view is used to save the current parameter set and to load previously stored ones, which allows a proven set of settings to be recalled for a known material. The *Back* button returns to the machine selection screen, and *Fullscreen* expands the software to the full display area.



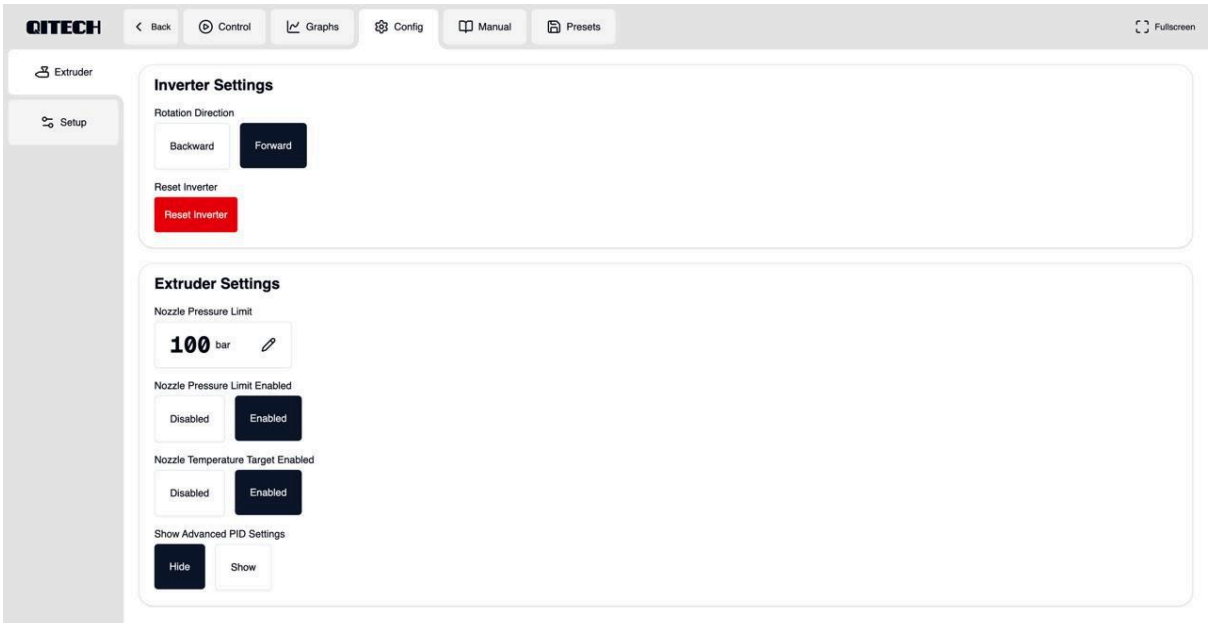
Control	
Heating Front / Heating Middle / Heating Back / Heating Nozzle	
Temperature	Measured actual temperature of the heating zone in °C, with live trend
Target Temperature	Editable setpoint for the heating zone in °C
Heating Power	Current electrical power output of the heater band in Watt
Screw Drive	
Status badge	Healthy / fault; Health status of the inverter and drive

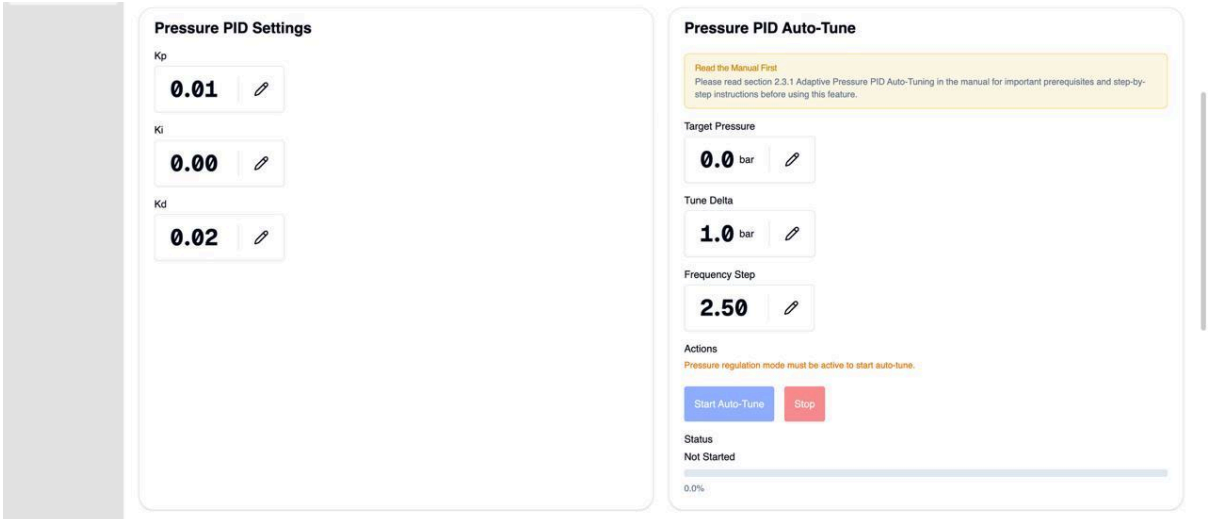
Regulation	Selects whether the screw speed follows a speed setpoint in RPM or a melt pressure setpoint in bar
Target Output RPM / Target Pressure	Screw speed setpoint, active in RPM mode / Melt pressure setpoint, active in Pressure mode
Rpm	Measured screw speed in rpm, with live trend
Pressure	Measured melt pressure at the die in bar, with live trend
Mode	
Standby	All heaters and the drive are switched off. The machine remains energised
Heat	Heating zones control to their setpoints. The screw drive stays locked
Extrude	Releases the screw drive. Only available once the heating zones have reached a minimum of 100 °C
Power Consumption	
Total Power	Total electrical power drawn by the machine in Watt
Motor Power	Electrical power drawn by the drive motor in Watt
Motor Current	Current drawn by the drive motor in Ampere
Total Energy Consumption	Cumulative energy consumed since the counter was last reset in kWh

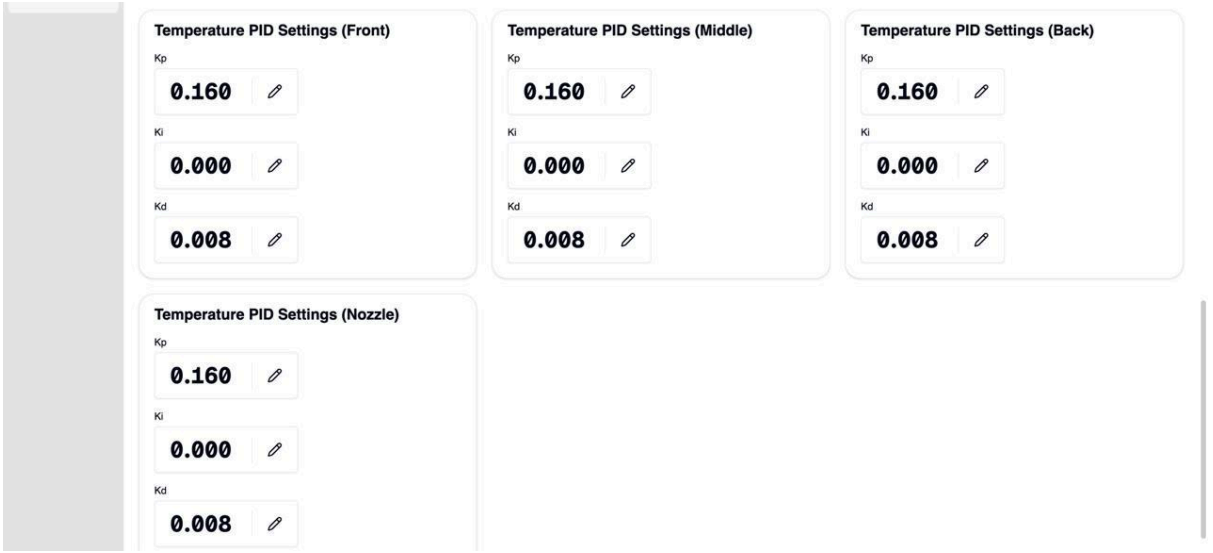


Graph	
Pressure	Melt pressure at the die in bar over time
Temperature	Actual temperatures of the Front, Middle, Back and Nozzle zones in °C over time
Power Outputs	Heating power of the Front, Middle, Back and Nozzle zones, motor power and total power in Watt over time
Motor current	Current drawn by the drive motor in Ampere over time

The graphs can be displayed either as a historical view or as a live view that updates continuously. A time range of up to one hour can be selected. Using the *Add Marker* and *Remove Marker* buttons, specific points in time can be marked manually, named and displayed in the graph. The *Export* button allows the process data to be exported in Excel format.

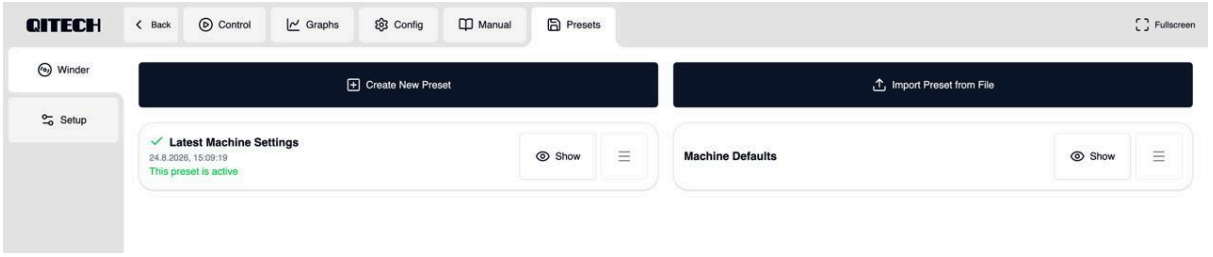

QITECH
Back
Control
Graphs
Config
Manual
Presets
Fullscreen
Extruder
Setup
Inverter Settings
Rotation Direction
Backward
Forward
Reset Inverter
Reset Inverter
Extruder Settings
Nozzle Pressure Limit
100 bar
Nozzle Pressure Limit Enabled
Disabled
Enabled
Nozzle Temperature Target Enabled
Disabled
Enabled
Show Advanced PID Settings
Hide
Show


Pressure PID Settings
Kp
0.01
Ki
0.00
Kd
0.02
Pressure PID Auto-Tune
Read the Manual First
Please read section 2.3.1 Adaptive Pressure PID Auto-Tuning in the manual for important prerequisites and step-by-step instructions before using this feature.
Target Pressure
0.0 bar
Tune Delta
1.0 bar
Frequency Step
2.50
Actions
Pressure regulation mode must be active to start auto-tune.
Start Auto-Tune
Stop
Status
Not Started
0.0%


Temperature PID Settings (Front)
Kp
0.160
Ki
0.000
Kd
0.008
Temperature PID Settings (Middle)
Kp
0.160
Ki
0.000
Kd
0.008
Temperature PID Settings (Back)
Kp
0.160
Ki
0.000
Kd
0.008
Temperature PID Settings (Nozzle)
Kp
0.160
Ki
0.000
Kd
0.008

Config	
Inverter Settings	
Rotation Direction	Direction of rotation of the screw, *Backward* or *Forward*. *Forward* is the normal extrusion direction
Reset Inverter	Acknowledges an inverter fault and resets the drive
Extruder Settings	
Nozzle Pressure Limit	Maximum permissible melt pressure at the die in bar. Default value is 100 bar
Nozzle Pressure Limit Enabled	Activates monitoring of the pressure limit. The screw drive is stopped when the limit is exceeded
Nozzle Temperature Target Enabled	Activates temperature control of the nozzle heating zone
Show Advanced PID Settings	Shows or hides the controller parameters described below
Pressure PID Settings	
Kp	Proportional gain of the pressure controller
Ki	Integral gain of the pressure controller
Kd	Derivative gain of the pressure controller
Pressure PID Auto-Tune	
Target Pressure	Pressure in bar around which the tuning routine oscillates
Tune Delta	Permissible pressure deviation in bar during the tuning routine
Frequency Step	Step size in Hz by which the inverter frequency is varied during tuning
Start Auto-Tune	Starts the tuning routine. Pressure regulation mode must be active
Stop	Aborts the tuning routine
Status	Text and progress bar showing the state of the tuning routine, for example *Not Started*
Temperature PID Settings	

Kp	Proportional gain of the zone temperature controller
Ki	Integral gain of the zone temperature controller
Kd	Derivative gain of the zone temperature controller



Presets	
Create New Preset	Saves the current machine configuration as a new named preset
Import Preset from File	Loads a previously exported preset from a file
Show	Displays the full configuration parameters of the selected preset
Menu (≡)	Opens additional options for the selected preset, such as renaming, exporting or deleting

1.4 Intended use

The QiTech Extruder is designed for melting thermoplastic granulate into filament with a diameter of 1–3 mm. It can be operated as a standalone machine or as part of a process line together with the QiTech Aquapath or QiTech Airpath and the QiTech Winder. In the process line, the machines are connected to one another by the power supply cables and LAN cables, with the QiTech Extruder as the lead machine of the line.

The machine must be placed on a stable, level surface with the parking brakes applied. The granulate is fed in through the hopper and conveyed by the screw through the heated barrel, where it is melted in several heating zones. The melt is pressed through the melting head and the nozzle to form the filament, which is then cooled and wound by the downstream machines of the line.

The screw is an exchangeable part of the machine. It may be exchanged to remove material that has adhered to the screw or has jammed in the barrel, and to fit a screw profile better

suited to the material being processed. Only screws with a screw geometry as defined in section 1.2 are permitted.

The following conditions of use are not permitted:

- Operation in potentially explosive atmospheres
- Operation in food production environments
- Operation outdoors or outside enclosed buildings
- Operation with screws that do not comply with the intended screw geometry
- Any use beyond the scope defined in these operating instructions

Other intended uses include setting up the machine and carrying out cleaning, maintenance, screw exchange, troubleshooting, fault rectification and repair work as described in these operating instructions. In the event of significant modifications of the machine or deviations from the intended use, a new risk assessment must be carried out and all changes must be documented accordingly.

1.5 Reasonably foreseeable misuse

The following types of use are considered reasonably foreseeable misuse and are not permitted:

Melting of unsuitable material:

- Feeding material that cannot be melted, or that is highly flammable or explosive
- Feeding material that releases toxic or corrosive substances when heated
- Feeding contaminated or insufficiently dried granulate
- Producing filament outside the permitted diameter range of 1–3 mm

Improper operation:

- Operating the extruder on an uneven surface, in any position other than the intended horizontal position, or on an unbraked trolley
- Starting extrusion before the machine has reached the required operating temperature
- Operating the heating elements without thermal mass
- Reaching into hot machine parts or into the hopper
- Dismantling machine parts that are surrounded by hardened plastic, in particular the melting head, screw and pressure sensor
- Use of spare parts or components not authorised by the manufacturer
- Operating the extruder unsupervised, or by tired or stressed personnel
- Removal or bypassing of safety guards or manipulation of sensors, or circumventing software restrictions

1.6 Significant modification / other use

Any structural, mechanical or electrical modification to the machine that has not been expressly authorised by QiTech Industries GmbH is considered a significant modification. Significant modifications may affect the safety of the machine, render the existing risk assessment invalid and void the CE conformity of the machine. In such cases, a new conformity assessment must be carried out before the machine is put back into service. Replacing the nozzle or the melting head with identical parts does not constitute a significant modification.

Any use of the machine beyond the scope defined in section 1.4 is considered improper use. QiTech Industries GmbH and its authorised representatives accept no liability for damage to persons or property resulting from improper use or unauthorised modifications. In such cases, liability rests solely with the operator.

2. Safety

2.1 General safety instructions

These safety instructions apply to all persons working on or with the QiTech Extruder. All personnel must be familiar with these instructions and with the information in the following chapters.

Personnel and authorisation

The machine may only be operated by professionally trained and instructed persons. Uninstructed persons and children must not have access to the machine. Repairs and maintenance work may only be carried out by qualified specialist personnel who are familiar with the relevant occupational safety regulations. Work on the electrical equipment of the machine may only be carried out by a qualified electrician.

Personal protective equipment

Appropriate personal protective equipment must be worn at all times. In particular:

- Always wear slip-resistant safety footwear
- Use a hairnet if you have long hair
- Do not wear loose clothing, ties, open shirt sleeves or jewellery
- Wear heat-resistant gloves when handling hot components such as the barrel, melting head or nozzle
- Hearing protection must be worn if the emission sound pressure level at the workplace exceeds 80 dB(A)

Electrical safety

Before carrying out any maintenance, cleaning or repair work, the machine must be switched off and fully de-energised. The following steps must be observed:

- Switch off the machine at the main switch and turn it to the "O" position
- Lock the main switch to secure the machine against unintentional restart
- Disconnect the power connection
- Have the machine de-energised by a qualified electrician
- Cover or isolate neighbouring live parts

The extruder is supplied with a three-phase CEE plug and must be connected to a suitable three-phase socket with a right-hand rotating field. The machine must be connected to a mains circuit protected by a residual current device (RCD). Before connection, have the socket outlet or connection box checked by a qualified electrician. Only use original fuses with the prescribed current ratings. Never use the machine with defective, crushed or bare cables.

Activating the emergency stop device shuts down the entire machine and disconnects it from the mains supply.

General operational safety

Before starting work, carry out a visual inspection to ensure the machine is in perfect technical condition. In particular, check that the die and melting head are fully screwed in. The machine may only be used when all protective devices are properly fitted and fully operational. Never leave the machine in operation without supervision, and stay with the machine until it has come to a standstill.

- Keep the work area clean and free of obstacles; do not place objects on the machine
- Switch off the machine when it is not in use
- Protect the electrical components from moisture
- Do not reach into moving parts of the machine or into the hopper
- Keep a safe distance from rotating parts
- Check the machine regularly for damage; only use original spare parts authorised by the manufacturer
- Ensure adequate lighting in accordance with applicable national workplace regulations

The barrel, melting head and nozzle reach high temperatures during operation. Do not touch hot parts and avoid contact with the molten material.

QiTech Industries GmbH accepts no liability for damage caused by failure to observe these operating instructions. The use of non-authorised spare parts or accessories will invalidate the guarantee and warranty.



Danger to life due to technically faulty machine
The machine may only be used when it is in perfect technical condition, in accordance with its intended use and in compliance with these operating instructions. All faults, especially those that could impair safety, must be rectified immediately before starting the machine.

2.2 Pictograms, symbols and warning notices

This chapter provides a comprehensive reference of all safety symbols that operators may encounter in an industrial working environment. Graphical symbols, including warning signs, mandatory signs, prohibition signs, follow ISO 7010. All people working with the QiTech Extruder must be familiar with the meanings of these symbols.

The general warning symbol can be used in combination with one of the following signal words:

DANGER Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.

WARNING Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.

NOTICE Indicates important information not related to personal injury. Non-observance may result in damage to property or the environment.

Warning signs		Mandatory signs	
	W001 General warning sign		M001 General warning sign
	W004 Warning of laser beam		M003 Use hearing protection
	W011 Warning of slippery surface		M004 Use eye protection
	W012 Warning of electrical voltage		M008 Use foot protection
	W016 Warning of toxic substances		M009 Use hand protection
	W017 Warning of hot surface		M010 Use protective clothing
	W018 Warning against automatic start-up		M012 Use handrail

	W019 Warning of crushing hazard		M014 Use head protection
	W024 Warning against hand injuries		M021 Switch off before maintenance or repair
	W025 Warning against counter-rotating rollers		
Prohibition signs		**Other pictograms**	
	P001 General prohibition sign		F001 Fire extinguisher
	P002 No smoking		WEEE symbol Separate collection of waste electrical and electronic equipment
	P003 No open flames; fire, open source of ignition and smoking not permitted		
	P022 Eating and drinking not permitted		

	P024 Entering the area not permitted		
	P031 Switching not permitted		
	D-P006 Access not permitted for unauthorised persons		

2.3 Safety devices

The QiTech Extruder is equipped with the following safety devices. All safety devices must be fully operational before the machine is started. It is not permitted to operate the machine with any safety device removed, bypassed or disabled. If a safety device is opened or removed for maintenance purposes, it must be correctly refitted and verified before the machine is switched on again.

Main switch
The machine has a main switch that connects and disconnects the mains supply of the QiTech Extruder or the whole process line. The main switch can be secured with a lock in the "O" position to prevent the machine from being switched on by unauthorised persons.

Emergency stop device
The emergency stop switch is used to bring the machine to a controlled standstill in the event of danger. Activating the emergency stop device shuts down the entire machine or process line and disconnects it from the mains supply. The device also prevents the machine from being switched on again after a power failure. The machine can only be restarted by physically resetting the emergency stop switch and then starting one of the operating modes in the software.

Overload protection
The inverter of the main motor is equipped with overload protection. It switches off the main drive motor automatically in the event of excessive power consumption or excessive temperature, by means of PTC thermistors in the motor coils. The motor can only be switched on again after the machine has been restarted.

Overheating protection
The heating is equipped with an overheating protection. If a heating zone exceeds its permissible operating temperature, the heating elements switch off automatically. A warning is issued in the event of sensor errors. In addition, the temperature that can be set by the operator is limited to the permissible range by the software. The heating should not be restarted until it has cooled sufficiently and the cause of overheating has been identified.

Undertemperature protection
The software only allows the main motor to be switched on once a temperature of 100 °C or higher has been reached.This minimises the risk of screw breakage or motor damage caused by starting up against excessive resistance.

Overpressure protection
The machine is equipped with overpressure protection in the software that switches off the main drive motor immediately in the event of unexpectedly high nozzle pressure. As an additional, mechanical safeguard, a rupture disc is fitted that opens at 350 bar and allows the melt to escape in a controlled manner, protecting the machine against damage caused by overpressure.

Notice: If safety-relevant changes or defects are detected on the machine, shut it down immediately, secure it against restart and report the issue to the responsible person before resuming operation.

2.4 Qualification of personnel / instruction and training

All persons working with the QiTech Extruder must have the qualifications appropriate to their assigned tasks. The following personnel categories are defined for the purposes of these operating instructions:

Operator: The operator is the person who uses, adjusts and monitors the machine in day-to-day operation. The operator is characterised by practical experience in handling the machine as well as in-depth knowledge of its functions and the correct course of action in the event of a fault.

Instructed person: An instructed person has been briefed by the operator on the tasks assigned to them and on the potential hazards associated with improper behaviour. The correct execution of all activities carried out by an instructed person is monitored by specialised personnel.

Specialised personnel: Specialised personnel are able to carry out their assigned tasks independently and to recognise and avoid potential hazards, based on their specialist training, knowledge and experience as well as their familiarity with the relevant regulations. It is assumed that specialised personnel have also received the instruction described above.

Qualified electrician: A qualified electrician is a person who, by virtue of their technical training, knowledge and experience as well as their familiarity with the relevant regulations,

is able to assess the work assigned to them and recognise and avoid potential hazards related to electrical systems.

Tasks	Minimum qualification
Transport	Specialised personnel of the manufacturer
Installation and commissioning	Specialised personnel of the manufacturer
Day-to-day operation	Instructed personnel of the operator
Setting, adjusting and restarting	Specialised personnel of the operator or manufacturer
Troubleshooting and fault rectification	Specialised personnel of the operator or manufacturer
Cleaning and maintenance	Specialised personnel of the operator or manufacturer
Decommissioning	Specialised personnel of the operator

As the operator, you are obliged to ensure that all personnel working on or with the machine are trained and instructed in accordance with these operating instructions before starting work. This includes instruction on all applicable legal and accident prevention regulations as well as on all safety devices on and around the machine. The varying technical qualifications of individual employees must be taken into account when planning instruction sessions.

Instruction must be understood and retained by all personnel. Compliance with instruction content must be verified regularly. Written confirmation of attendance must be obtained from each employee following each instruction session.

Restrictions on employment

The following persons must not work on or with the machine under any circumstances:

- Persons with skin diseases
- Persons with chronic respiratory conditions or a tendency to allergic reactions
- Pregnant women and nursing mothers
- Persons under 18 years of age

2.5 Residual risks

Despite the safety measures integrated into the design and construction of the QiTech Extruder, certain residual risks cannot be fully eliminated. The following hazards remain and must be taken into account by all persons working on or with the machine.



Danger! Operation only permitted with all protective devices fitted
The machine must only be operated with all protective devices correctly installed and fully functional. The operator is responsible for ensuring this before each use.



Danger! Drawing-in by moving parts
The rotating screw and other moving parts can draw in limbs, hair, loose clothing or jewellery. Do not reach into the hopper or into moving parts. Wear close-fitting clothing, use a hairnet for long hair, and do not wear gloves during activities with a risk of being drawn in.



Danger! Hot surfaces and molten material
The barrel, heating zones, melting head, nozzle and the emerging filament reach high temperatures during operation and remain hot for some time after switch-off. Avoid contact with hot parts and with the molten material. Wear heat-resistant gloves when working on hot components.



Danger! Hot melt ejected when the rupture disc opens
If the pressure exceeds the response value, the rupture disc on the side of the melting head opens and hot melt is ejected under pressure to the side. Keep clear of the side of the melting head during operation and do not position yourself in the ejection path. After the rupture disc has opened, allow the machine to cool and de-energise it before any intervention.



Danger! Vapours and emissions from the melt
Heating the material can release vapours or fumes. Ensure sufficient ventilation of the installation site; an extraction system at the nozzle is recommended. Do not process material that releases toxic or corrosive substances when heated.



Danger! Electric shock
Before carrying out any repair, set-up or maintenance work, the machine must be fully de-energised. Do not resume work until the absence of voltage has been confirmed.

3. Transport, installation and commissioning

3.1 Transport

Warning! Risk of injury due to improper transport
When loading and unloading the machine, do not place any limbs under machine parts when setting them down. Do not stand under suspended loads or between moving loads and walls or other machine parts. Wear appropriate personal protective equipment during all transport operations.

The QiTech Extruder is delivered fully assembled on its mobile trolley. Before transport, the following steps must be carried out:

- Switch off the process line at the main switch on the QiTech Extruder and disconnect the mains plug
- Allow the machine to cool completely before moving it
- Disconnect and secure all cables and accessories to prevent damage during transport
- Ensure the parking brakes of the trolley are released before moving the machine

The machine may be moved manually on its castors over short distances on a level surface. For longer distances or loading onto a vehicle, suitable lifting equipment must be used. The permissible load-bearing capacity of the lifting equipment must meet the machine weight. After transport, inspect the machine for any damage that may have occurred during transit.

Caution! Risk of damage due to incorrect installation and commissioning
Errors during installation and commissioning can cause considerable damage to property. Installation and commissioning may only be carried out by the manufacturer's employees.

3.2 Requirements for the installation site

Before installing the machine, ensure that the installation site meets the following requirements:

- The surface must be level, solid and capable of supporting the weight of the machine
- The location must be used exclusively for industrial or commercial purposes and located indoors
- Sufficient ventilation of the installation site must be ensured; an extraction system at the nozzle is recommended
- The location must be as fire-free as possible and equipped with a fire extinguisher
- Sufficient clearance must be maintained around the machine to allow safe operation and unobstructed access for maintenance work
- Adequate lighting must be provided in accordance with applicable national workplace regulations

3.3 Basic installation

Caution! Risk of damage due to incorrect installation and commissioning
Errors during installation and commissioning can cause considerable damage to property. Installation and commissioning may only be carried out by the manufacturer's employees.

The QiTech Extruder is delivered fully assembled on its mobile trolley with the latest available version of the QiTech Control software installed. After positioning the machine at the intended installation site, carry out the following steps:

- Position the machine at the intended installation site in accordance with the requirements
- Apply the parking brakes of the trolley to secure the machine against unintentional movement
- Connect the power supply cable from the QiTech Extruder to the QiTech Aquapath, and the remaining power supply cables if operated as the whole process line
- Connect the LAN cables: the "Out" port of each machine is connected to the "In" port of the next machine. The ports are labelled accordingly.
- Route all cables in a way that they cannot be tripped over

Calibration of the temperature and pressure sensors
The sensors are pre-calibrated ex works. Manual recalibration is only possible for the pressure sensor. Please refer to the sensor manufacturer's instructions for this.

3.4 Electrical connection

Danger! Electric shock due to improper electrical connection
All work on the electrical connection must be carried out by a qualified electrician. Ensure the mains socket is de-energised before connecting the machine.

Before connecting the machine to the mains supply, carry out the following steps:

- Verify that the mains voltage and frequency at the installation site correspond to the specification of the machine
- Connect the three-phase CEE plug of the extruder to a suitable three-phase socket with a right-hand rotating field. If only a left-hand rotating field is available, the direction of rotation can alternatively be adjusted in the "Setup" tab of the software; the correct direction of rotation is marked with an arrow on the extruder
- Ensure that the mains socket is equipped with an earthing contact and has been checked by a qualified electrician
- Ensure that the mains circuit is protected by a residual current device (RCD)
- Only use original fuses with the prescribed current ratings to avoid overloading

3.5 Commissioning

Before switching on the machine for the first time, carry out a full visual inspection:

- All protective devices are correctly fitted
- All cables are undamaged and correctly routed
- The parking brakes of the trolley are applied
- The die and the melting head are fully screwed in
- The emergency stop device is not actuated

Once all checks have been completed satisfactorily, the machine may be switched on.

4. Operation

4.1 Starting the machine

A visual inspection must be carried out every working day before the machine is started. In particular, check that the die and the melting head are fully screwed in, that all protective devices are correctly fitted, all cables are undamaged, and the machine shows no signs of damage.

- Make sure that the emergency stop device is not actuated
- Switch on the process line at the main switch on the QiTech Extruder
- Turn on the display using the button on the right-hand side of the display
- The QiTech Extruder will appear in the left column of the QiTech Control software

4.2 Preheating the extruder

The extruder must reach its operating temperature before extrusion can begin. The main motor can only be switched on once a temperature of 100 °C or higher has been reached.

- Select *Control* in the function bar of the QiTech Control software
- Press the *Heat* button in the *Mode* box
- Set the target temperature of the four heating zones according to the material being processed
- Set the target temperature of the nozzle
- After the target temperatures have been reached, wait approximately 5 minutes until the material in the screw barrel has completely melted

Only sufficiently dried granulate may be used. Residual moisture in the granulate impairs the filament quality and can lead to defects in the extruded strand. Observe the drying instructions for the material being processed.

Danger! Hot surfaces and molten material
The barrel, melting head, nozzle and the emerging filament reach high temperatures. Avoid contact with hot parts and with the molten material.

4.3 Extrusion

The QiTech Extruder can be operated in two regulation modes. In speed regulation, the screw runs at a fixed set speed. In pressure regulation, the screw speed is adjusted automatically to keep the nozzle pressure at the set target value. The regulation mode is selected on the *Control* page.

Starting the extrusion:

- Fill the dried granulate into the hopper
- In speed regulation, set the screw speed via the *Screw Speed* slider; in pressure regulation, set the target pressure
- Press the *Extrude* button in the *Mode* box
- Wait until the old melt has been flushed out of the extruder, i.e. the filament is no longer discoloured and a constant material flow has been achieved

Once a constant strand is being produced, the filament is fed into the downstream cooling unit by hand. Use a tool such as pliers to guide the hot strand, and do not touch it directly.

Adjusting the process variables during operation

- Adjust the desired parameters using the input fields
- Press the *Heat* or *Extrude* function button to accept the parameters

Changing the material during operation

- Empty the current material from the hopper
- To flush the extruder screw and nozzle, cleaning granulate may be fed in until the emerging material runs clear
- Preheat to the temperature of the new material
- Fill the new material into the hopper
- Restart the extrusion and wait until any contamination from the previous material has completely disappeared

4.4 Stopping and switching off

To shut down the machine:

- Press the *Standby* button in the QiTech Control software; the screw and the heating elements are switched off
- Switch off the process line at the main switch on the QiTech Extruder and turn it to the "O" position

In case of emergency, the machine can be shut down immediately using the emergency stop device.

If the machine is not to be used for a prolonged period, disconnect the mains plug.

Notice: If safety-relevant changes or defects are detected during operation, shut down the machine immediately with the main switch on the QiTech Extruder and report the issue to the responsible person before resuming operation.

5. Maintenance and inspection

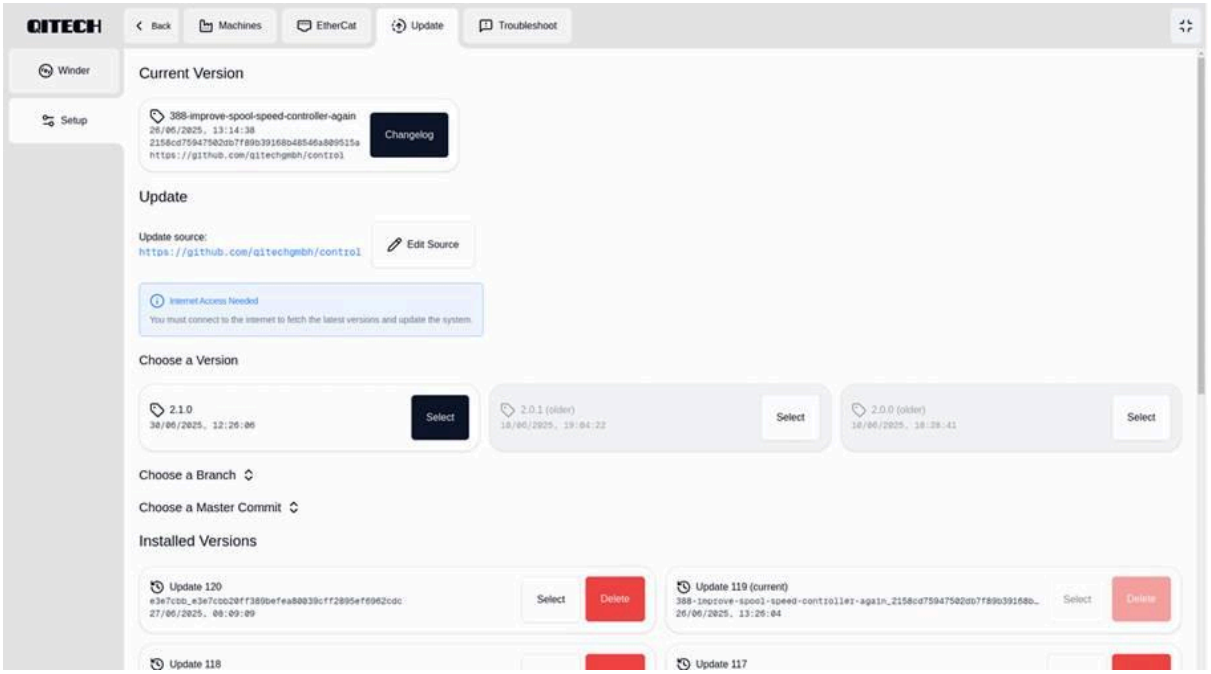
5.1 Updating the QiTech Control software

Software updates may contain bug fixes, new functions or security improvements. It is therefore recommended to check regularly whether updates are available and to install them promptly.

To perform a software update, proceed as follows:

- Ensure the machine has an active internet connection
- Open the QiTech Control software and navigate to *Setup* and *Updates*
- Available updates will be displayed; select the desired update
- The display will shut down and restart automatically; the updated software version will be installed upon restart

Previous software versions and the complete update history can also be viewed under *Setup* and *Updates*.



Notice: Modifying the software source or installing unofficial software versions may compromise the safety and functionality of the machine and will invalidate the guarantee and warranty. Only official updates from QiTech GmbH may be installed.

5.2 Mechanical maintenance

Maintenance and repair work on the machine may only be carried out by a trained person. Heat-resistant gloves must be worn when working on hot components.

All work described in this section should be carried out while the machine is warm. Residual material only remains soft while the machine is at temperature, and components can be released easier and without damage in this condition. Heat the machine up to operating temperature first, then switch off the main switch.



Warning! Burns.
The nozzle head, the nozzle plate, the barrel, the screw and the residual melt remain hot after switching off the machine. Wear heat-resistant gloves and eye protection, avoid any contact with molten material and keep the working area clear.



Warning! Unexpected start-up.
Before removing any component, switch the main switch on the QiTech Extruder to "0" and secure it against being switched on again. In a process line, the main switch de-energises the entire line. Never reach into the open barrel while the drive is energised.

5.2.1 Replacing the nozzle plate

The nozzle plate can be replaced in the event of excessive wear, a change in the required throughput, or to achieve greater back pressure when processing composites. Nozzle plates with any nozzle diameter up to a maximum of 3 mm may be used. Proceed as follows:

1. Preheat the system to operating temperature.
2. Switch the main switch to "0" and secure it against being switched on again.
3. Remove the nozzle plate using an allen key
4. Clean the nozzle plate with a brass brush while it is still warm, as plastic residues from extrusion adhere to it. Do not use steel tools, as these damage the surfaces. A hot air gun can be used to soften stubborn residues.
5. Insert the cleaned or new nozzle plate and secure it using a minimum tightening torque of 25 Nm.
6. Retighten the screws while the nozzle plate is hot, as a slight loss of preload may occur due to thermal effects.

5.2.2 Replacing the burst disc

The burst disc is located on the nozzle head and protects the machine against excessive melt pressure. It is only replaced after it has responded. A burst disc that has responded is recognised by molten material escaping from it. Proceed as follows:

1. Preheat the system to operating temperature.
2. Switch the main switch to "0" and secure it against being switched on again.
3. Unscrew the burst disc from the nozzle head with a screwdriver.

4. Clean the seat in the nozzle head while it is still warm. A hot air gun is a useful aid for removing material residues.
5. Screw in a new burst disc

A burst disc that has responded must always be discarded and must never be refitted. The technical drawing of the rupture disc can be found in Annex B: Technical Drawing of the Nozzle, D (2:1).

5.2.3 Cleaning or replacing the sieve

The sieve is located behind the nozzle plate and retains contaminants in the melt. It can be cleaned and reused or replaced with a new sieve. Proceed as follows:

1. Preheat the system to operating temperature.
2. Switch the main switch to "0" and secure it against being switched on again.
3. Remove the nozzle plate using an allen key
4. Remove the sieve and breaker plate. Both components are secured with a retaining ring, which must first be removed using the appropriate tool, such as retaining-ring pliers.
5. Insert a new sieve and refit the nozzle plate.

5.2.4 Screw exchange

The screw can be exchanged if material has fused onto it, if the screw is blocked, or if a screw profile better suited to the material is to be fitted. Only screws with a screw geometry as defined in section 1.2 are permitted. The threaded rod required for the procedure is stored on the back of the extruder.

Disassembly

1. Empty the hopper and flush the machine until no residue of the previous material is discharged.
2. Set the machine to standby mode. The machine must remain warm to keep the residual material soft.
3. Set the main switch to the “0” position and secure it against being switched on again.
4. Remove the nozzle head. The nozzle head is removed as one assembly, consisting of the nozzle plate, pressure sensor, temperature sensor, and nozzle heating band. Keep all cables connected.
5. Place the nozzle head in the nozzle-head holder on the side of the machine and secure it with the supplied nut and bolt.

Check that the keyway of the gearbox bushing is aligned in the lower half. This alignment ensures that the key does not fall out of the keyway when the screw is pushed out. If the keyway is not facing downward, remove the motor from the gearbox and turn the motor shaft by hand until the keyway is facing downward.

6. Take the threaded rod from its holder on the side of the extruder and insert it into the cylinder from the rear.

7. Turn the threaded rod in evenly using a screwdriver to push the screw forward and out of the cylinder. Do not use impact tools.
8. Support the screw as it emerges from the front and place it in a heat-resistant container. The screw is hot and heavy.
9. Remove the threaded rod and return it to its holder on the side of the extruder.

Installation

1. Clean the screw and the cylinder bore while they are still warm using a brass brush. Do not use steel tools, as they may damage the surfaces.
2. Inspect the screw for wear, scoring, and coating damage. Do not reinstall a damaged screw.

Before installing the new screw, check whether the key is still located in the keyway of the gearbox shaft. If so, align the keyway of the screw with the existing key before inserting the screw. When inserting the screw, ensure that the key and the corresponding keyway of the gearbox shaft are correctly aligned.

3. Insert the new or cleaned screw into the cylinder from the front and push it in until it engages with the key in the bearing housing. Engagement should be clearly perceptible. The front end of the screw at the extruder outlet must be positioned slightly below the flange surface on which the nozzle is mounted. If necessary, carefully drive the screw further into the extruder using controlled blows with a soft-faced hammer until the required position is reached.
4. Reinstall the nozzle head. Always use new screws for installation and fit the specified wedge-lock washers. Tighten the screws to a minimum torque of 25 Nm.
5. After the machine has been heated up for the first time, retighten the screws to compensate for any loss of preload caused by thermal expansion and settling.
6. Check the Nozzle Pressure Limit in the Config view. The pressure PID and temperature PIDs may need to be readjusted.
7. Extrude a test piece and check the filament diameter and melt quality before resuming production.

6. Troubleshooting

The following table provides an overview of the most common faults, their possible causes and recommended corrective actions. If a fault cannot be resolved using the measures described below, contact QiTech Industries GmbH for technical support.

Before attempting to rectify any fault, ensure that the machine is in a safe condition. For faults requiring access to internal components, switch off the machine and disconnect the main plug.

Fault	Possible cause	Corrective action
Machine is not recognised by Control software	LAN cable not connected	Check and reconnect the LAN cable
	Outdated software version	Update the QiTech Control software
	Network configuration error	Check network settings
Main motor does not start	Emergency stop device actuated	Reset the emergency stop device and restart the mode in the software
	Operating temperature not yet reached	Preheat the extruder; the motor is only released at 100 °C or higher
	Overload protection triggered	Restart the machine; check for excessive load or blockage; check if screw is blocked
	Mains plug not connected	Check the mains connection and the power supply
Temperature sensors issue warnings	Loose contact or cable breakage	Check the connections between the temperature measuring unit and the control unit

	Motor and sensor cables routed in parallel	Route the motor cable and sensor cables as far apart as possible; never in parallel
Melt does not have the desired consistency or colour	Temperature set too high	Reduce the temperature; check whether the material has decomposed or burnt
	Temperature set too low, or preheating insufficient	Increase the temperature and ensure sufficient preheating
	Residual old material in the screw	Flush the screw with the new material until no mixture remains
	Granulate not dried, or starting material contaminated	Use dried, uncontaminated granulate
Material throughput is not constant	Material in the hopper bridged or fused together	Check and loosen the material in the hopper
	Granulate grains too large to be drawn in	Use granulate of a suitable grain size
	Melting head or nozzle blocked	Check and clean the melting head or nozzle
	Insufficient material in the hopper	Ensure there is always enough material in the hopper
Deposits form on the melting head	Temperature at the melting head too low	Increase the temperature of the melting head or the front heating zone
	Insufficient insulation	Additionally insulate the melting head
Nozzle pressure too high	Temperature too low / melt too viscous	Increase the temperature; reduce the screw speed

	Nozzle blocked	Depressurise, check the nozzle and restart only once the pressure has dropped
Screw is blocked	Operating temperature not yet reached, material not fully melted	Preheat all heating zones to their target temperature before starting extrusion
	Material fused onto the screw after a long dwell time at temperature	Set the rotation direction to Backward briefly to relieve the screw; if the screw remains blocked, take it out for cleaning and exchange if needed
	Screw geometry not suited to the material being processed	Fit a suitable screw; only screws with a screw geometry as defined in section 1.2 are permitted

7. Decommissioning, disassembly and disposal

When the machine has reached the end of its planned service life, QiTech Industries GmbH recommends contacting the manufacturer to arrange for the machine to be taken back. If this is not possible, the machine must be permanently taken out of service and disposed in accordance with the instructions in this chapter and in compliance with applicable national regulations.

Disassembly and decommissioning may only be carried out by trained and qualified specialist personnel of the operator.

Decommissioning

Before disassembly begins, the machine must be taken out of service by properly switching off, disconnecting the power supply and securing the machine against unintentional restart. Allow the machine to cool completely before proceeding with disassembly.

Disassembly

Disassembly requires an M5 Allen key and a standard screwdriver set. During disassembly, wear safety footwear, protective gloves and protective clothing.

The machine consists largely of aluminium and plastic components, which can be easily separated by loosening the screw connections. Proceed as follows:

- Remove all cables and set aside for separate disposal
- Remove the aluminium composite panels from the housing frame by loosening the screw connections
- Separate the mechanical components from the electronic components
- Remove and separately collect all auxiliary materials

Disposal

All components must be separated by material class to enable subsequent recycling or differentiated scrapping. The following materials require separate disposal by an authorised specialist company:

- Electrical and electronic components
- Lubricants and oils

Equipment bearing the WEEE symbol must not be disposed of as unsorted waste. The operator is responsible for ensuring that all waste is separated by material and disposed of via an authorised collection or take-back point in accordance with applicable national regulations.

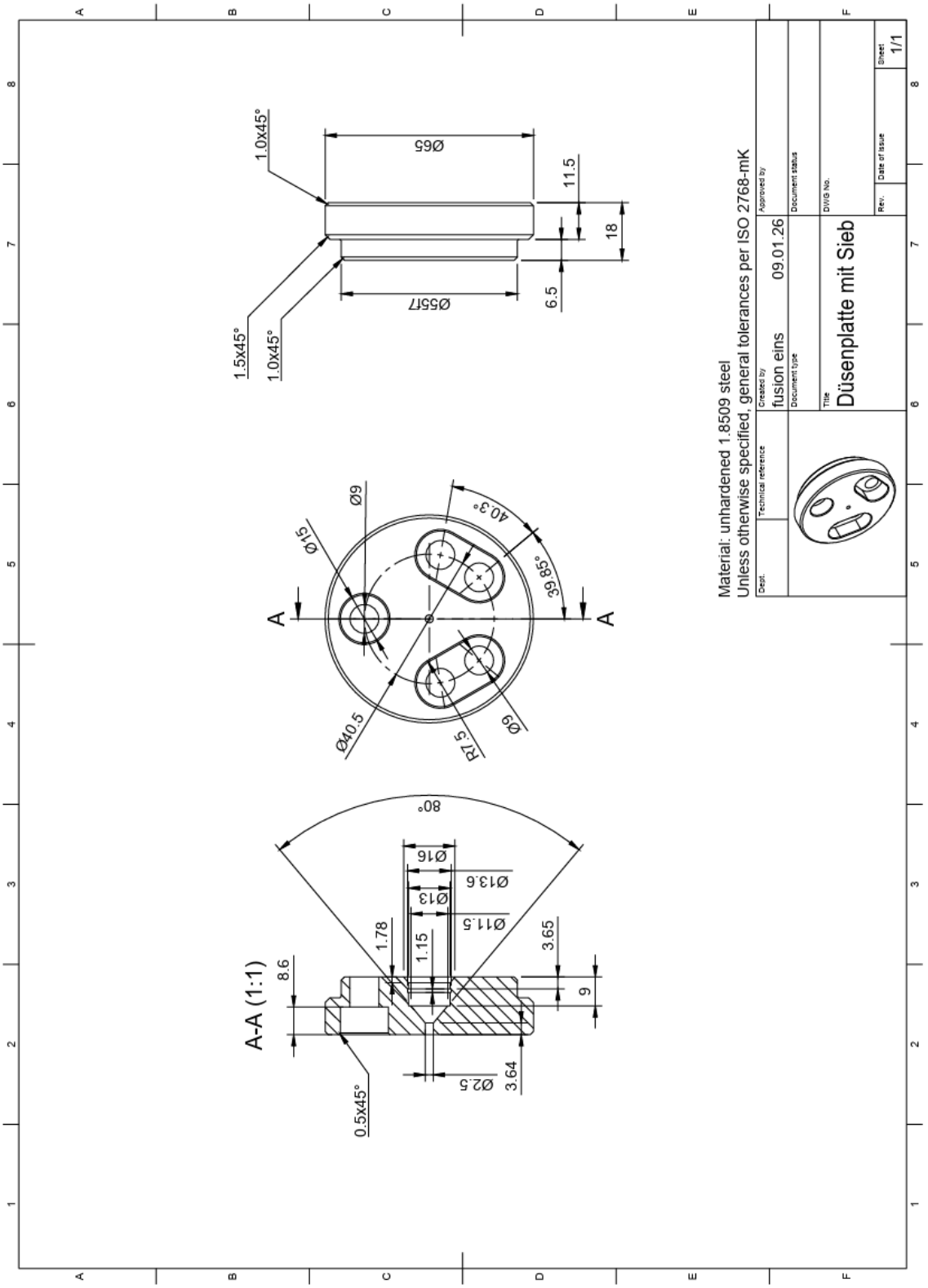
Notice: Disposal of hazardous materials such as lubricants, oils and electronic components must be carried out in accordance with applicable national environmental regulations. Contact an authorised waste disposal specialist if in doubt.

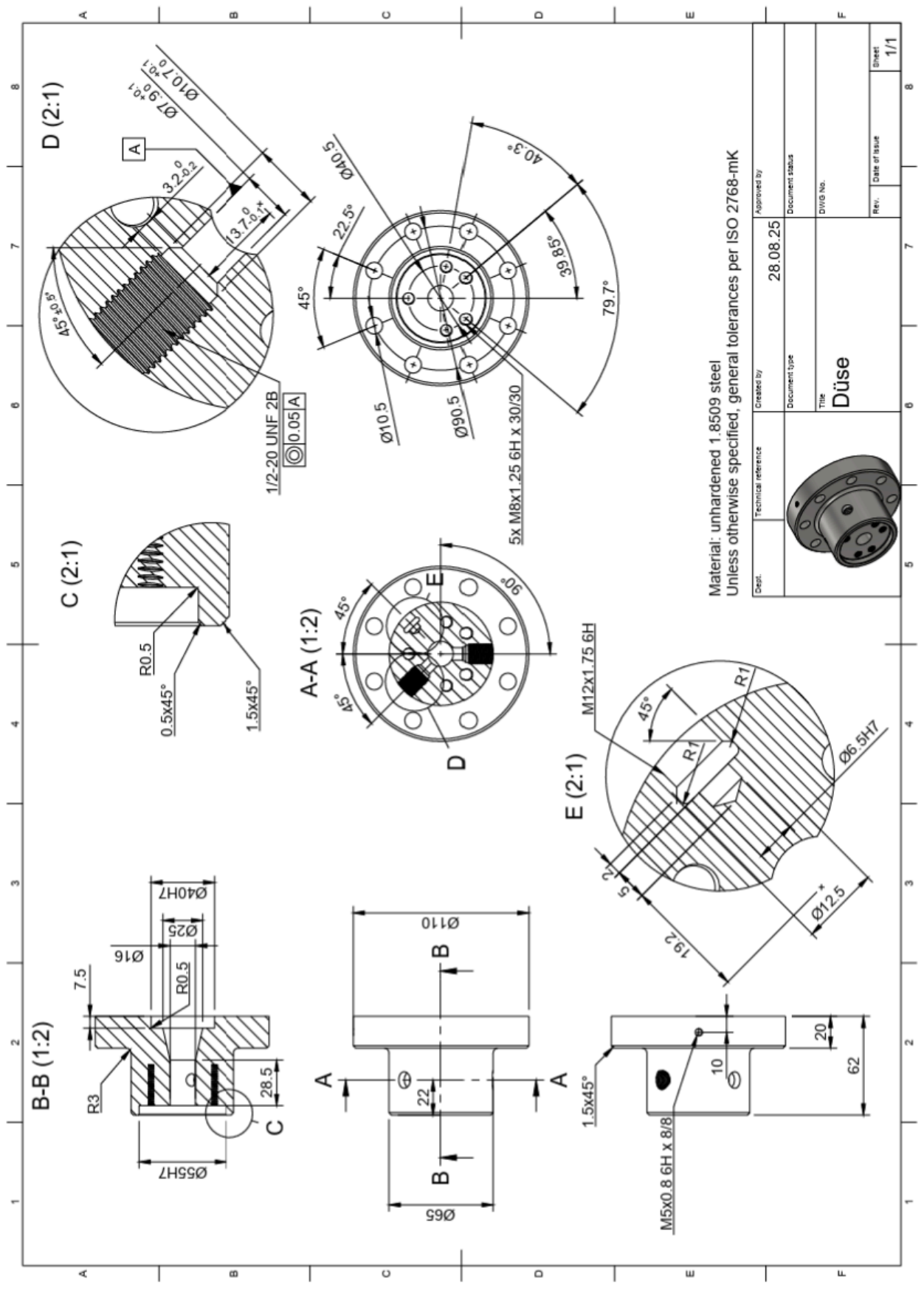
8. Annexes

Annex A: Rendering of the process line

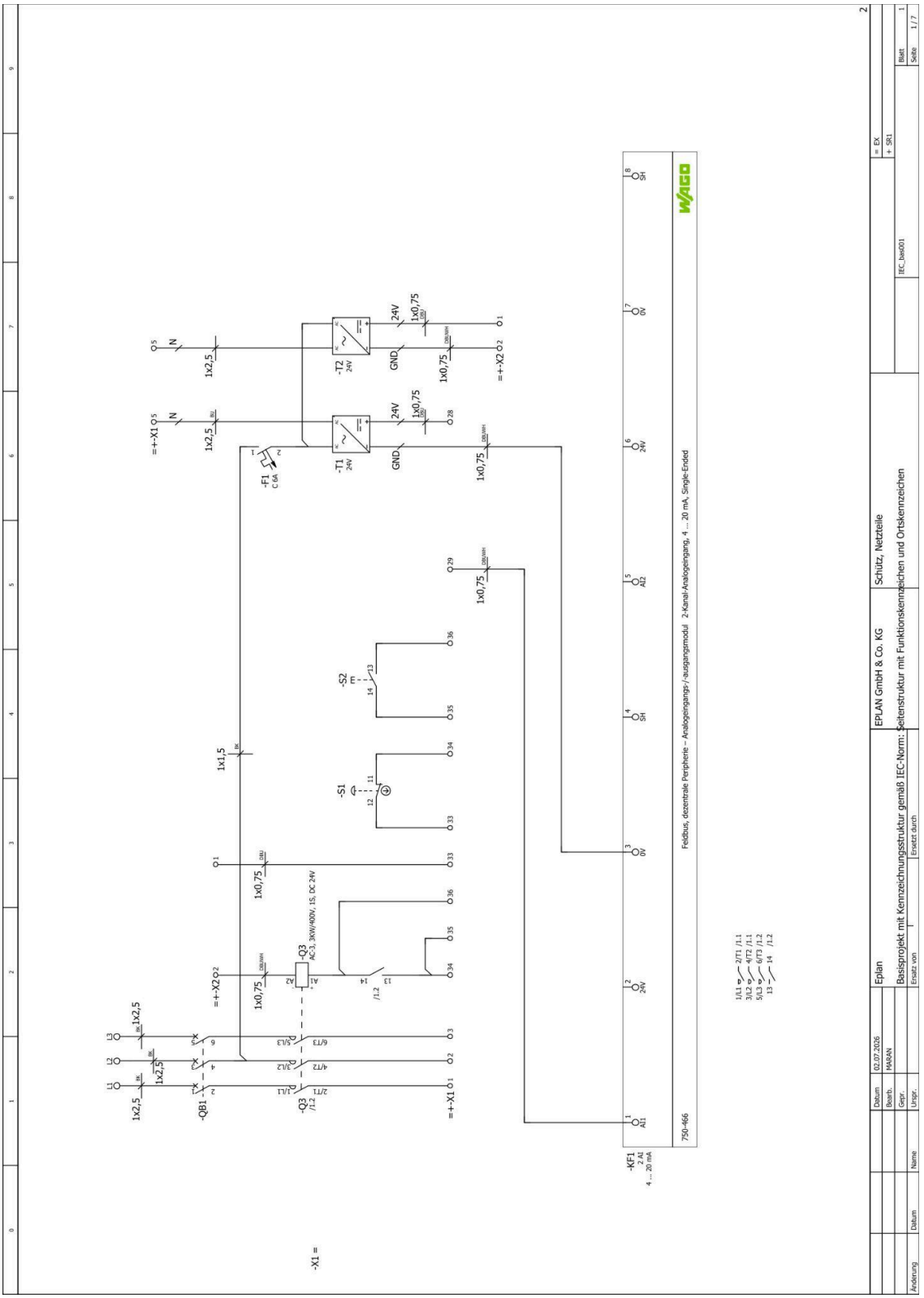


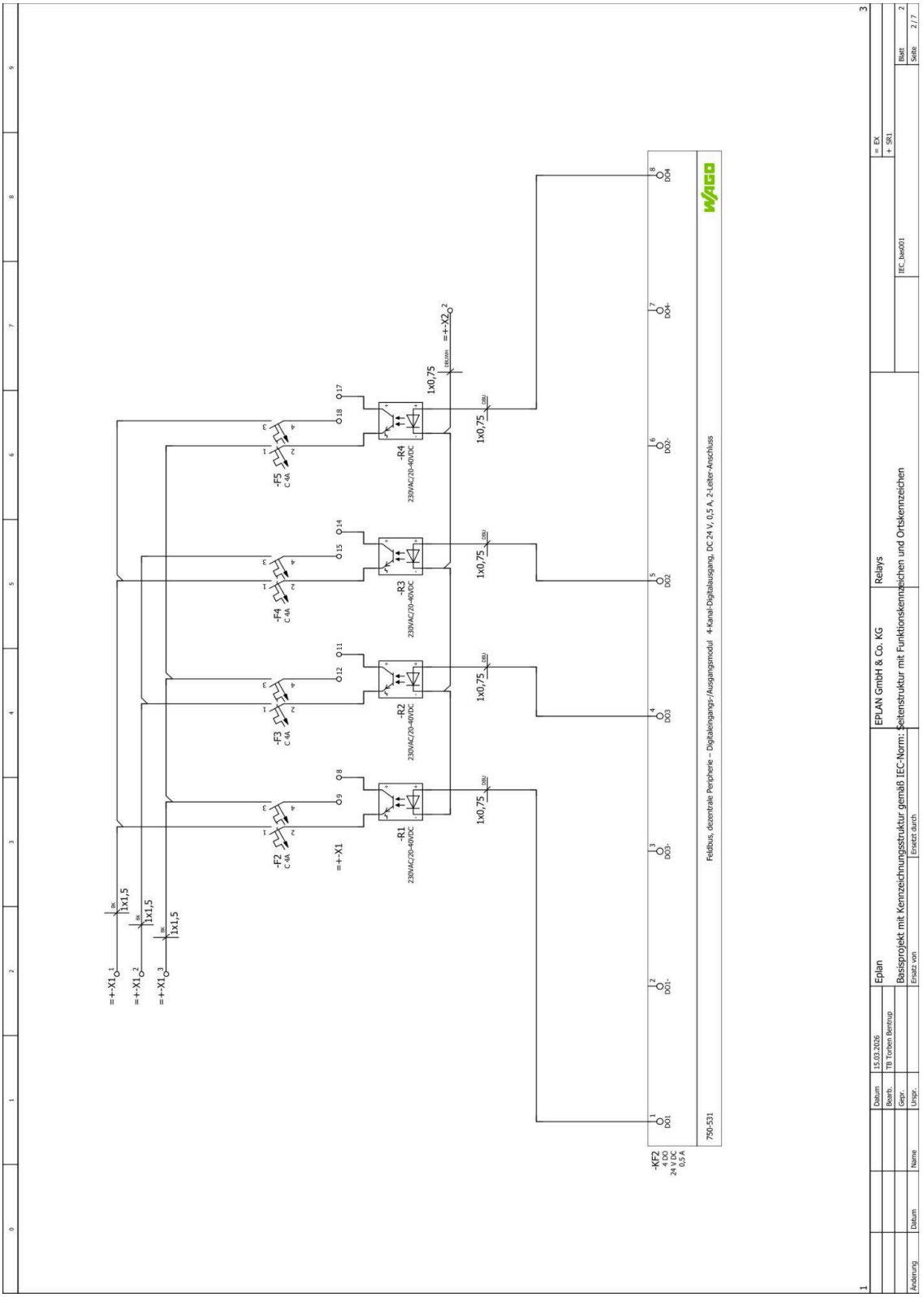
Annex B: Technical drawing of the nozzle

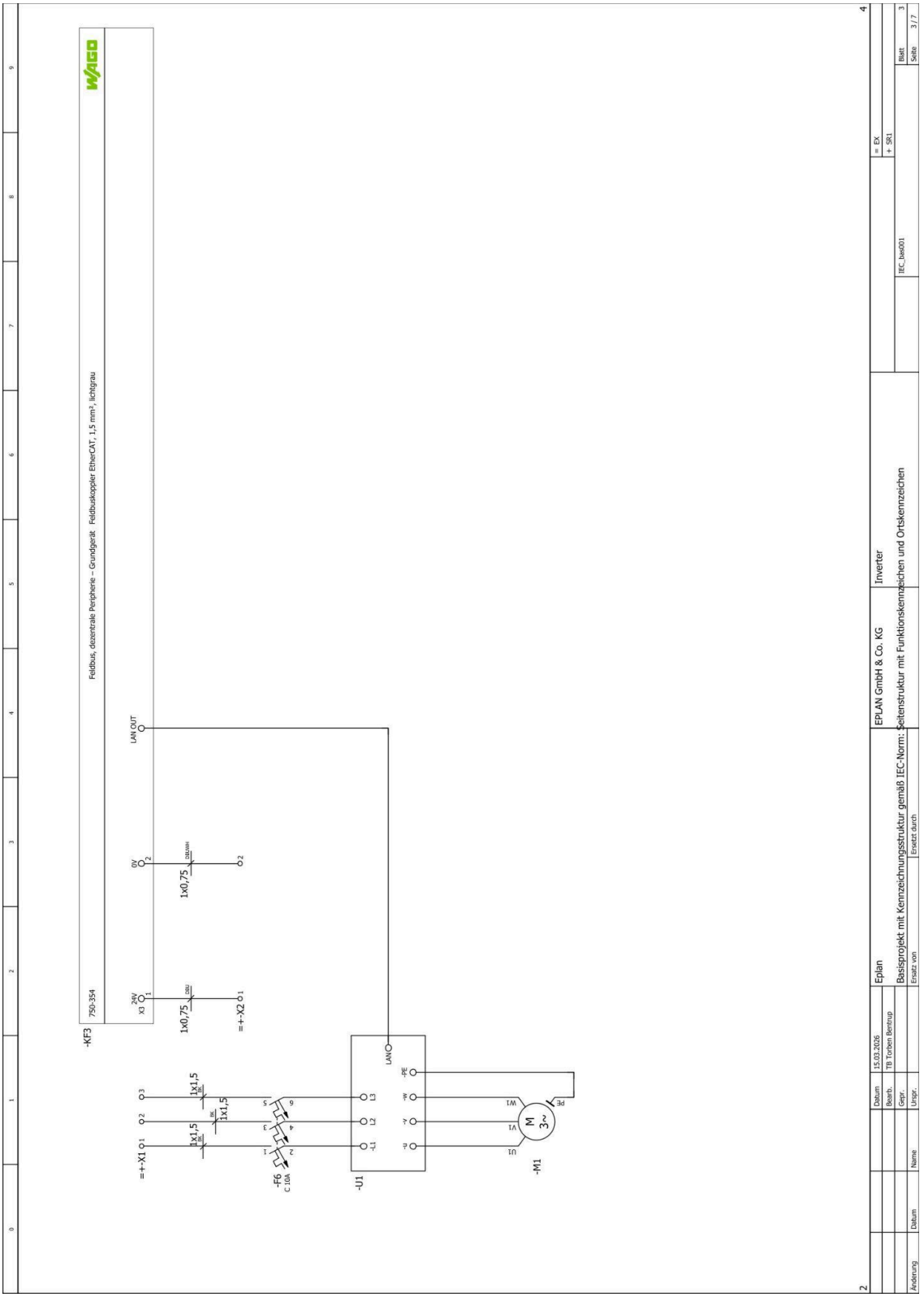


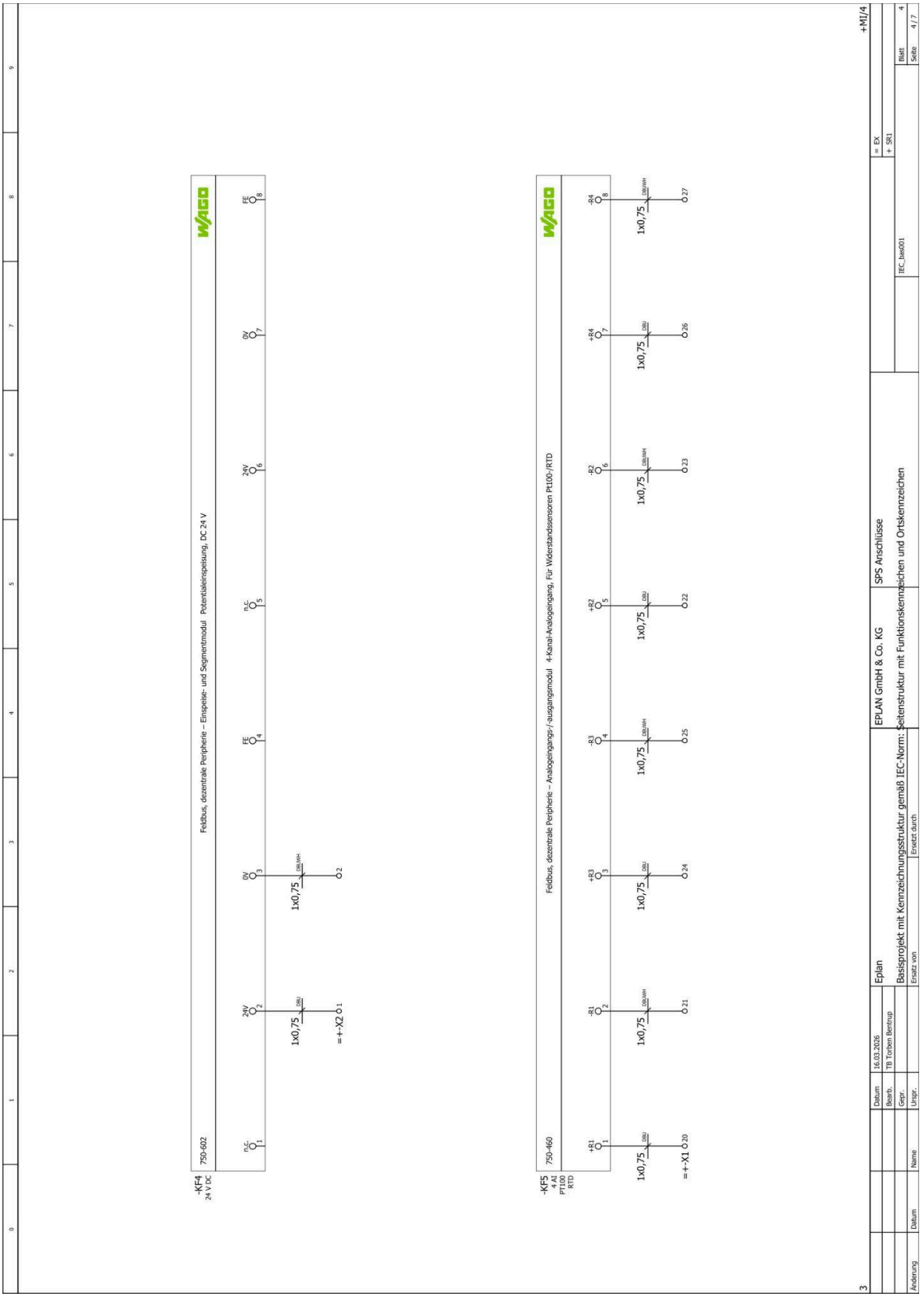

D (2:1)
C (2:1)
A-A (1:2)
B-B (1:2)
E (2:1)
1/2-20 UNF 2B
0.05 A
5x M8x1.25 6H x 30/30
M12x1.75 6H
M5x0.8 6H x 8/8
Material: unhardened 1.8509 steel
Unless otherwise specified, general tolerances per ISO 2768-mK
Dept.
Technical reference
Created by
28.08.25
Approved by
Document type
Document status
Title
Düse
DWG No.
Rev.
Date of issue
Sheet
1/1

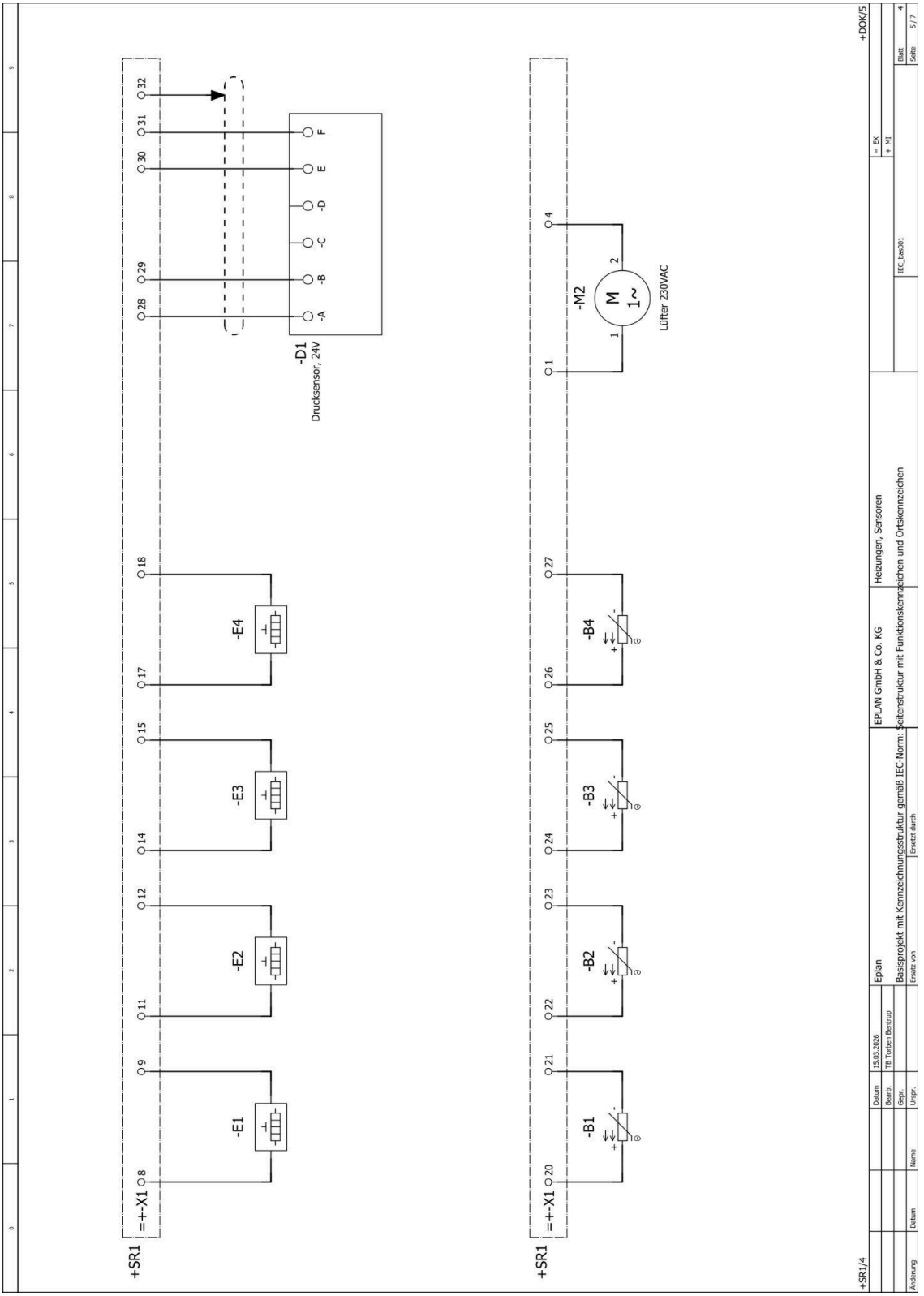
Annex C: Electrical documentation and parts list




=+-X1
=+-X2
-F2
-F3
-F4
-F5
-R1
-R2
-R3
-R4
-KF2
750-531
Feldbus, dezentrale Peripherie – Digitaleingangs-/Ausgangsmodul 4-Kanal-Digitalausgang, DC 24 V, 0,5 A, 2-Leiter-Anschluss
EPLAN GmbH & Co. KG
Relays
Eplan
Basisprojekt mit Kennzeichnungsstruktur gemäß IEC-Norm: Seitenstruktur mit Funktionskennzeichen und Ortskennzeichen
15.03.2026
TB Torben Bentrup
IEC_basis01
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-KF3
750-354
Feldbus, dezentrale Peripherie – Grundgerät Feldbuskoppler EtherCAT, 1,5 mm², lichtgrau
LAN OUT
0V
24V
1x0,75
=+-X2
=+-X1
1x1,5
-F6
C 10A
-U1
LAN
-M1
M
3~
EPLAN GmbH & Co. KG
Inverter
Eplan
Basisprojekt mit Kennzeichnungsstruktur gemäß IEC-Norm: Seitenstruktur mit Funktionskennzeichen und Ortskennzeichen
15.03.2026
TB Torben Bentrup
IEC_basic01
Blatt 3
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-KF4
24 V DC
750-602
Feldbus, dezentrale Peripherie – Einspeise- und Segmentmodul Potentialeinspeisung, DC 24 V
WAGO
24V
0V
FE
1x0,75
=+-X2
-KF5
4 AI
PT100
RTD
750-460
Feldbus, dezentrale Peripherie – Analogeingangs-/-ausgangsmodul 4-Kanal-Analogeingang, Für Widerstandssensoren Pt100-/RTD
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EPLAN GmbH & Co. KG
SPS Anschlüsse
Basisprojekt mit Kennzeichnungsstruktur gemäß IEC-Norm: Seitenstruktur mit Funktionskennzeichen und Ortskennzeichen
Eplan
TB Torben Bentrup
16.03.2026
+M1/4
= EX
+ SR1
IEC_basis01
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+SR1
=+-X1
-E1
-E2
-E3
-E4
-D1
Drucksensor, 24V
-A
-B
-C
-D
-E
-F
-M2
M
1~
Lüfter 230VAC
-B1
-B2
-B3
-B4
EPLAN GmbH & Co. KG
Heizungen, Sensoren
Basisprojekt mit Kennzeichnungsstruktur gemäß IEC-Norm: Seitenstruktur mit Funktionskennzeichen und Ortskennzeichen
+DOK/5
+SR1/4

Artikelstückliste

F01_001

Betriebsmittelkennzeichen	Menge	Bezeichnung	Typnummer	Lieferant	Artikelnummer
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-X1	3	Durchgangsreihenklemme	2202-1404		WAGO.2202-1404
-X1	1	Schutzleiter-Reihenklemme	2202-1407		WAGO.2202-1407
-X1	24	Durchgangsreihenklemme	2202-1201		WAGO.2202-1201
-X1	6	Schutzleiter-Reihenklemme	2202-1207		WAGO.2202-1207
-X2	4	Durchgangsreihenklemme	2202-1201		WAGO.2202-1201
+SR1-F1	1	Sicherungsautomat - S200 - 1P - C - 6 A	S201-C6		ABB.2CDS251001R0064
+SR1-F2	1	Sicherungsautomat - S200 - 2P - C - 4 A	S202-C4		ABB.2CDS252001R0044
+SR1-F3	1	Sicherungsautomat - S200 - 2P - C - 4 A	S202-C4		ABB.2CDS252001R0044
+SR1-F4	1	Sicherungsautomat - S200 - 2P - C - 4 A	S202-C4		ABB.2CDS252001R0044
+SR1-F5	1	Sicherungsautomat - S200 - 2P - C - 4 A	S202-C4		ABB.2CDS252001R0044
+SR1-F6	1	Sicherungsautomat - S200 - 3P - C - 10 A	S203-C10		ABB.2CDS253001R0104
+SR1-KF1	1	Feldbus, dezentrale Peripherie – Analogeingangs-/-ausgangsmodul	750-466		WAGO.750-466
+SR1-KF2	1	Feldbus, dezentrale Peripherie – Digitaleingangs-/Ausgangsmodul	750-531		WAGO.750-531
+SR1-KF3	1	Feldbus, dezentrale Peripherie – Grundgerät	750-354		WAGO.750-354
+SR1-KF4	1	Feldbus, dezentrale Peripherie – Einspeise- und Segmentmodul	750-602		WAGO.750-602
+SR1-KF5	1	Feldbus, dezentrale Peripherie – Analogeingangs-/-ausgangsmodul	750-460		WAGO.750-460
+SR1-M1	0				
+SR1-Q2	0				
+SR1-Q3	1	SCHUETZ,AC3:3KW 1S DC24V,STEH	3RT2015-1BB41-1AA0	SIE	SIE.3RT2015-1BB41-1AA0
+SR1-R1	0				
+SR1-R2	0				
+SR1-R3	0				
+SR1-R4	0				
+SR1-S1	0				
+SR1-S2	0				
+SR1-T1	1	<<???>>	HDR-15-5		MWP.HDR-15-5
+SR1-T2	1	<<???>>	HDR-60-24		MWP.HDR-60-24
+SR1-U1	0				
+SR1-W1	1	Multi-Standard SC 2.2 1X2,5 BK	Multi-Standard SC 2.2	LAPP	LAPP.4150501
+SR1-W2	1	Multi-Standard SC 2.2 1X2,5 BK	Multi-Standard SC 2.2	LAPP	LAPP.4150501
+SR1-W3	1	Multi-Standard SC 2.2 1X2,5 BK	Multi-Standard SC 2.2	LAPP	LAPP.4150501
+SR1-W4	1	Multi-Standard SC 2.2 1X1,5 BK	Multi-Standard SC 2.2	LAPP	LAPP.4150401
+SR1-W5	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W6	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W7	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W8	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W9	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W10	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W11	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W12	1	Multi-Standard SC 2.2 1X1,5 BK	Multi-Standard SC 2.2	LAPP	LAPP.4150401
+SR1-W13	1	Multi-Standard SC 2.2 1X1,5 BK	Multi-Standard SC 2.2	LAPP	LAPP.4150401
+SR1-W14	1	Multi-Standard SC 2.2 1X1,5 BK	Multi-Standard SC 2.2	LAPP	LAPP.4150401
+SR1-W15	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W16	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W17	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W18	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W19	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W20	1	Multi-Standard SC 2.2 1X1,5 BK	Multi-Standard SC 2.2	LAPP	LAPP.4150401
+SR1-W21	1	Multi-Standard SC 2.2 1X1,5 BK	Multi-Standard SC 2.2	LAPP	LAPP.4150401
+SR1-W22	1	Multi-Standard SC 2.2 1X1,5 BK	Multi-Standard SC 2.2	LAPP	LAPP.4150401
+SR1-W23	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W24	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W25	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W26	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W27	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K

+MI/4

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			Datum	16.03.2026	Eplan	EPLAN GmbH & Co. KG	Artikelstückliste : WAGO.2202-1401 - LAPP.4510142K	= EX	
			Bearb.	TB Torben Bentrup				+ DOK	
			Gepr.		Basisprojekt mit Kennzeichnungsstruktur gemäß IEC-Norm: Seitenstruktur mit Funktionskennzeichen und Ortskennzeichen			IEC_bas001	Blatt 5
Änderung	Datum	Name	Urspr.		Ersatz von	Ersetzt durch			Seite 6 / 7

Artikelstückliste

F01_001

Betriebsmittelkennzeichen	Menge	Bezeichnung	Typnummer	Lieferant	Artikelnummer
+SR1-W28	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W29	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W30	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W31	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W32	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W33	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510142K
+SR1-W34	1	Anschluss- und Steuerleitungen	H05V-K in big one-way cardboard box		LAPP.4510922K
+SR1-W35	1	Anschluss- und Steuerleitungen	H07V-K		LAPP.4520022
+SR1-W36	1	Anschluss- und Steuerleitungen	H07V-K		LAPP.4520022
+MI-B1	0				
+MI-B2	0				
+MI-B3	0				
+MI-B4	0				
+MI-D1	0				
+MI-E1	0				
+MI-E2	0				
+MI-E3	0				
+MI-E4	0				
+MI-M2	0				

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			Datum	16.03.2026	Eplan	EPLAN GmbH & Co. KG	Artikelstückliste : LAPP.4510922K -		= EX
			Bearb.	TB Torben Bentrup					+ DOK
			Gepr.		Basisprojekt mit Kennzeichnungsstruktur gemäß IEC-Norm: Seitenstruktur mit Funktionskennzeichen und Ortskennzeichen			IEC_bas001	Blatt 5.1
Änderung	Datum	Name	Urspr.		Ersatz von	Ersetzt durch			Seite 7 / 7